NMRobotics_Getting_Started

Getting Started with NMRobotics

Local AI setup guide for Coglet · Ubuntu 22.04 / 24.04 · Estimated setup: 30–60 minutes

NMRobotics connects your Coglet to three AI services running on your computer. Once models are downloaded, speech, reasoning and voice synthesis can run locally without cloud AI services.

Service	Purpose	Recommended
LLM	Reasoning, replies, tool calls	Ollama + Gemma 4 E4B
ASR	Speech to text	Speaches + faster-whisper
TTS	Text to speech	Kokoro-FastAPI

0. System requirements

Ubuntu 22.04 or 24.04; 16 GB RAM or more; approximately 20 GB free disk space; an NVIDIA GPU with 8 GB VRAM is recommended for the LLM. Coglet and the computer must be on the same local network. CPU-only operation may be slower.

Proxy note: Clash/Mihomo fake-IP or TUN configurations can cause model downloads to resolve to 198.18.x.x. Add +.ollama.com, +.ollama.ai, +.r2.cloudflarestorage.com, +.huggingface.co and +.hf.co to the proxy fake-ip-filter, then restart it.

1. Install NMRobotics

Download the GitHub ZIP, extract it, open a terminal in the extracted folder and run:

```
./install.sh
nmrobotics help
```

The installer prepares ffmpeg, libopus0, python3-venv and a private .venv environment. If ~/.local/bin is missing from PATH, log out and back in.

2. Start the server and console

```
nmrobotics start
nmrobotics open
```

The console opens at http://127.0.0.1:18002/. On Overview, note the OTA URL, such as http://192.168.1.50:18000/nm/ota/. Your LAN address will differ.

3. Install local AI services

3.1 Ollama and Gemma 4

```
curl -fsSL https://ollama.com/install.sh | sh
ollama pull gemma4:e4b
```

Increase the context window so Coglet’s system instructions and tool definitions are not truncated. Use 8192 instead if 16384 is too demanding for your hardware.

```
sudo mkdir -p /etc/systemd/system/ollama.service.d
printf '[Service]\nEnvironment="OLLAMA_CONTEXT_LENGTH=16384"\n' | sudo tee
/etc/systemd/system/ollama.service.d/override.conf
sudo systemctl daemon-reload
sudo systemctl restart ollama
```

3.2 Docker

```
sudo apt update
sudo apt install -y docker.io
sudo usermod -aG docker $USER
```

Log out and back in to activate Docker group membership. The examples below use CPU ASR and TTS to reserve GPU memory for the LLM.

3.3 Kokoro-FastAPI (TTS)

```
docker run -d --name kokoro --restart unless-stopped \
  -p 127.0.0.1:8880:8880 \
  ghcr.io/remsky/kokoro-fastapi-cpu:latest
curl http://127.0.0.1:8880/v1/audio/voices
```

3.4 Speaches (ASR)

```
docker run -d --name speaches --restart unless-stopped \
  -p 127.0.0.1:8000:8000 \
  -e WHISPER__INFERENCE_DEVICE=cpu \
  -e WHISPER__TTL=-1 \
  -v hf-hub-cache:/home/ubuntu/.cache/huggingface/hub \
  ghcr.io/speaches-ai/speaches:latest-cpu
curl -fsS -X POST http://127.0.0.1:8000/v1/models/Systran/faster-whisper-small
```

Use the multilingual Systran/faster-whisper-small model, not an English-only .en model. TTL=-1 prevents idle unloading, but does not guarantee protection against loading or inference hangs. Validate real transcription before relying on the service.

4. Configure AI Services in the console

Setting	Value
LLM Base URL	http://127.0.0.1:11434/v1
LLM Model	gemma4:e4b
LLM Context Window	16384 (or 8192)
Vision	Use the LLM above
ASR Base URL	http://127.0.0.1:8000/v1

ASR Model	Systran/faster-whisper-small
TTS Base URL	http://127.0.0.1:8880/v1
TTS Model / format	kokoro / pcm
Voices	en: af_heart; zh: zf_xiaobei; ja: jf_alpha; es: ef_dora

Leave API keys blank for unauthenticated local services. Click Save, then Overview → Test all. Confirm LLM tool calling, TTS audio, ASR transcription and Vision image understanding. The first request may take longer while loading a model.

5. Connect Coglet

Copy the OTA URL from Overview. Enter it through the firmware’ s Wi-Fi setup advanced options (if available) or configure CONFIG_OTA_URL with idf.py menuconfig, rebuild and flash. Restart Coglet; it should appear under Connected devices. No activation code is required.

```
sudo ufw allow 18000/tcp
```

Coglet recognizes wake words on-device. In Assistant → Wake words, enter the exact phrases supported by the installed firmware, one per line (for example “Hello Coglet” or “你好小智”). Activity displays “wake word: …” when matched; “device cue: …” may indicate a missing configured phrase.

Say	Expected action
Hello	Spoken response and matching eye emotion
Move your ears	Ear movement
What do you see?	Take a photo and describe it
Remind me in two minutes to drink water	Create a reminder
Bye	End conversation and return to standby

6. Everyday commands

Command	Action
nmrobotics status	Service status and OTA URL
nmrobotics restart	Restart background service
nmrobotics stop	Stop background service
nmrobotics logs	Live server logs
nmrobotics check	Check configured AI services
nmrobotics ports	Check ports 18000 and 18002

The Activity page records turns, tool calls, errors and timing. Settings are stored in data/config.yaml, which may contain credentials; do not share it publicly.

7. Update NMRobotics

Download a new release and copy updated application files over the existing installation, preserving data/ and .venv/. Back up your configuration first.

```
nmrobotics restart
# If requirements.txt changed:
./install.sh
```

8. Troubleshooting

Symptom	Check / fix
Ports 18000/18002 in use or console 403	Identify the listening process; remove obsolete containers only after confirming ownership.
Console unreachable	nmrobotics status, then nmrobotics logs.
Download resolves to 198.18.x.x	Review proxy fake-IP filter.

Speech works but actions do not	Verify LLM tool calling, context window and Activity device replies.
Wake word ignored	Check firmware wake word and Assistant settings.
Device not connected	Verify LAN, OTA URL and firewall TCP 18000.
ASR hangs	Check Docker logs, actual transcription and model loading; CPU-only systems can also hang.
TTS hangs	Check Kokoro logs and system resources.
Reminders at wrong time	Set time zone under Network & Time; restart Coglet.

```
docker ps
docker logs --tail 100 speaches
docker logs --tail 100 kokoro
docker stats --no-stream
curl http://127.0.0.1:8000/health
```

A successful ASR /health response does not prove the model is ready. Test a real WAV file:

```
curl --max-time 60 http://127.0.0.1:8000/v1/audio/transcriptions \
  -F "file=@test.wav" \
  -F "model=Systran/faster-whisper-small"
```

9. Uninstall

```
nmrobotics uninstall
rm ~/.local/bin/nmrobotics
# Optional: remove the AI containers
docker rm -f kokoro speaches
```

Delete the NMRobotics folder when no longer needed. Docker cache volumes and Ollama are managed separately.

Repository placement

Save this guide as docs/GETTING_STARTED.md in the repository. Add this line near the top of README.md:

```
New here? Start with [Getting Started](docs/GETTING_STARTED.md).
```

Release note: Kokoro multilingual synthesis has been tested; Speaches transcription has worked but reproducible timeouts have also been observed. Long-term ASR stability and version-specific commands should be verified before publishing this as a fully validated setup guide.